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Category

Lesson 5: Operations with Fractions

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Dividing Fractions

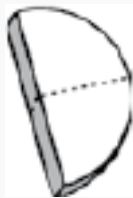
Division and multiplication are opposite operations. This important fact allows you to change division into multiplication.

To divide a fraction, change the division symbol to multiplication and invert the number you are dividing by. The inverted fraction is the reciprocal of the original fraction.

Example 1

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Angela has to give her cat $\frac{1}{2}$ of a pill twice a day. If she has $\frac{1}{4}$ of a pill left, how many more times can she give her cat its medicine?



Step 1
Invert the divisor (the number you are dividing *by*).

$\frac{1}{2} \div \frac{1}{4}$ $\frac{1}{4}$ inverts to $\frac{4}{1}$ $\frac{1}{2} \times \frac{4}{1} = \frac{4}{2} = 2$

Step 2
Change $\div$ into $\times$. Use canceling, if possible.

$\frac{1}{\cancel{2}} \times \frac{\cancel{4}^2}{1} = 2$

Step 3
Multiply across using the new numerators and denominators.

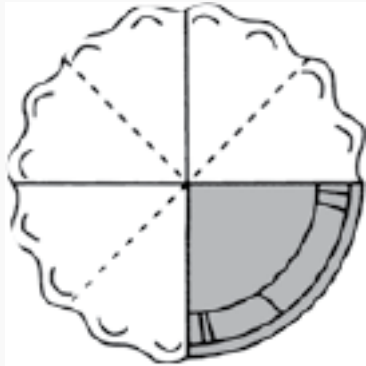
$\frac{1}{1} \times \frac{2}{1} = 2$

Angela can give her cat $\frac{1}{4}$ of a pill **2 more times**.

Example 2

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The Morrison family has $\frac{3}{4}$ of a pie left over. If there are 6 people in the family, what fraction of the original pie will each person get?



Step 1
Write the whole number as a fraction with a denominator of 1. Then invert the fraction.

$\frac{3}{4} \div \frac{6}{1}$ $\frac{6}{1}$ inverts to $\frac{1}{6}$ $\frac{3}{4} \times \frac{1}{6} = \frac{3}{24} = \frac{1}{8}$

Step 2
Change $\div$ into $\times$. Use canceling, if possible.

$\frac{\cancel{3}}{4} \times \frac{1}{\cancel{6}_2} = \frac{1}{8}$

Step 3
Multiply across using the new numerators and denominators.

$\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$

Each person will get $\frac{1}{8}$ of the pie.

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